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Research Article

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Prediction of the global occurrence of maize diseases and estimation of yield loss under climate change

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Abstract

BACKGROUND: Climate change significantly impacts global maize production via yield reduction, posing a threat to global food security. Disease-related crop damage reduces quality and yield and results in economic losses. However, the occurrence of diseases caused by climate change, and thus crop yield loss, has not been given much attention.

RESULTS: This study aims to investigate the potential impact of six major diseases on maize yield loss over the next 20 to 80 years under climate change. To this end, the Maximum Entropy model was implemented, based on Coupled Model Intercomparison Project 6 data. The results indicated that temperature and precipitation are identified as primary limiting factors for disease onset. Southern corn rust was projected to be the most severe disease in the future; with a few of the combined occurrence of all the selected diseases covered in this study were predicted to progressively worsen over time. Yield losses caused by diseases varied per continent, with North America facing the highest loss, followed by Asia, South America, Europe, Africa, and Oceania.

CONCLUSION: This study provides a basis for regional projections and global control of maize diseases under future climate conditions.
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Supporting information may be found in the online version of this article.

Keywords: climate change; CMIP6; MaxEnt model; maize diseases; prevalence prediction; yield loss; food security

ABBREVIATIONS

AMT	Annual mean temperature
AP	Annual precipitation
AR6	Sixth Assessment Report
AUC	Area under the curve
CA	Corn anthracnose
CCR	Common corn rust
CMIP	Coupled Model Intercomparison Project
FAO	Food and Agriculture Organization of the United Nations
GBIF	Global biodiversity information platform
GCMs	Global climatic models
IPCC	Intergovernmental Panel on Climate Change
IS	Isothermality
MAWM	Maximum temperature of the warmest month
MaxEnt	Maximum Entropy model
MDR	Mean diurnal range
MER	Maize ear rot
MICM	Minimum temperature of the coldest month
NLB	Northern leaf blight of corn
PCQ	Precipitation of the coldest quarter
PDM	Precipitation of driest month
PDQ	Precipitation of the driest quarter
PS	Precipitation seasonality
PWAQ	Precipitation of the warmest quarter
PWEQ	Precipitation of the wettest quarter
PWM	Precipitation of wettest month
ROC	Receiver operating characteristic
RCPs	Representative concentration pathways
SCR	Southern corn rust

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SLB	Southern leaf blight of corn
SSPs	Shared socioeconomic pathways
TAR	Temperature annual range
TCQ	Mean temperature of the coldest quarter
TDQ	Mean temperature of the driest quarter
TS	Temperature seasonality
TWAQ	Mean temperature of the warmest quarter
TWEQ	Mean temperature of the wettest quarter

1 INTRODUCTION

Climate change has resulted in negative impacts on ecosystems, biodiversity, and species distribution globally.[1] Climate change significantly affects agriculture and food security as temperatures rise and atmospheric carbon dioxide (CO_2) levels increase.[2] While the effect vary, climate change impacts the availability of agricultural resources, production processes, agricultural planting systems, and crop growth, thereby reducing productivity and food quality.[3] Therefore, it is crucial to understand the specific impact of climate change on disease occurrence leading to changes in crop yields in order to maintain food security.

Environmental conditions affected by climate such as temperature and extreme weather events play a crucial role in the interaction between host plants and pathogens for many different crops during disease development.[4–6] For example, elevated temperatures increase the possibility of the occurrence of rice blast epidemics, while an increase in CO_2 concentration causes rice leaf blast and sheath blight more serious infections, leading to negative impacts on rice yields. Likewise, the severity of Septoria leaf spot and stem rust in oat varieties is elevated due to their sensitivity to temperature.[7,8] Extreme weather events, such as floods, droughts, heat waves, and storms, can also worsen plant diseases and reduce production. For example, floods facilitate the spread of water-borne pathogens. Droughts and heat waves make host plants susceptible to infection,[9] while storms can enhance the airflow transmission of spores, threatening the physiological tolerance of plants, ultimately resulting in their death or large-scale reduction of production.[9–11] Future climate change is expected to exacerbate these challenges, creating new patterns in regions that have not encountered them previously.[12,13]

Maize (*Zea mays* L.), is one of the most extensively cultivated and highest-yielding food crops worldwide. As one of the most crucial crops globally, the total yield of maize is the highest of all crops. Total production reached 1.17×10^9 t in 2022 (https://www.fao.org/). On the one hand, as demand for maize grows, especially in sub-Saharan Africa where maize provides more than 50% of the calorie intake for millions of people,[14] the amount of maize planted is increasing, and the disease is also increasing. On the other hand, climate change is not optimistic, successive years of cultivation and climatic maladaptation have led to an increase in maize diseases.[15,16] Some of the most prevalent diseases include northern leaf blight of corn (NLB), maize ear rot (MER), and southern corn rust (SCR). For instance, NLB, which is one of the most devastating leaf diseases of maize, severely impacts maize production.[17,18] In a major NLB outbreak, the yield loss of maize can be up to 50%.[19,20] The pathogens of MER not only cause the disease but also produce a variety of fungal toxins, leading to losses of up to 70% of the expected yield in Chongqing and surrounding areas in China, and becoming a significant obstacle to maize quality and yield.[21,22] The occurrence of SCR has also caused severe yield losses in many countries, with maize yields in China being reduced by 42–53% during primary outbreak.[23,24] It is essential to predict the potential occurrence of maize diseases under future climatic conditions and estimate the yield losses caused by them so that targeted management measures can be taken to prevent a continuous increase in losses due to climate change.

Currently, various models have been employed to predict the occurrence of different crop yields and diseases. For instance, the Cropping System Model was utilized to investigate the negative effects of temperatures and reduced rainfall on rice yields via the Crop Estimation through Resource and Environment Synthesis rice crop estimation.[25] Previous study combined satellite and climate data to construct a prediction model that effectively projected wheat yield performance in Australia.[26] A species distribution model that incorporated climate data was employed to evaluate the potential impact of increased greenhouse gas emissions on the climatic suitability of maize cultivation.[27] Researchers at Sam Houston State University used the Maximum Entropy (MaxEnt) model to predict suitable habitats for pests of maize in the United States, identifying high-risk areas for planting maize.[28]

Climate change is an important factor affecting the occurrence of diseases. Changes in the occurrence of diseases directly affect crop yield and quality. Although some studies have explored the impact of single climatic factors such as rainfall,[29] high temperature[30] and drought stress,[31] on maize yield, few have predicted the future trends of significant maize diseases. However, the adverse impact of maize diseases on yields is a critical issue that cannot be underestimated. With the aggravation of maize diseases, the yield of maize in the field is reduced or even zero.[32,33] The MaxEnt model integrates machine learning and maximum entropy principles to simulate the potential geographic distribution of species based on layers of environmental variables and species distribution records.[34] It is a species distribution prediction model that has been used in epidemiological studies due to it being a flexible model whose results are easily interpretable. What sets it apart is that it can model sample-only data. Significantly, it can model complex non-linear relationships between response variables and predictors. The support variables are diverse, making it ideal for different research objectives and prediction needs.[35–39] Therefore, this study analyzes the spatial patterns and future changes of major maize diseases globally under various future climate scenarios using the MaxEnt model.

In this study, the major climatic variables contributing to disease variation were identified, and the proportion of each kind of disease occurrence level was calculated. The change of disease severity in the future was obtained, and the yield variation caused by disease occurrence was analyzed. The results of this study provide valuable insights into how the scope of maize diseases may expand or contract during future climate change. These findings serve as a basis for potential global partnership in controlling major diseases that undermine maize yield and quality while averting the expansion of disease range. Furthermore, this study informs the adoption of prompt disease management strategies by predicting the future occurrence of maize diseases.

2 MATERIALS AND METHODS

2.1 Data sources

Disease distribution data for this study were obtained from the Global Biodiversity Information Platform (GBIF, http://www.gbif.org, accessed on 10 October 2022), an integrated database of global biodiversity information. Six kinds of diseases that significantly reduce maize yield and are widely distributed around the

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world were selected as the focus of this study[40] (Fig. 1), using the Latin name of each disease as a keyword to search and download data from the GBIF platform. After excluding duplicate values and removing records lacking detailed longitude or latitude information, a total of 757 distribution points were used for MaxEnt model prediction, based on 1285 geographic coordinates. To calculate yield losses caused by diseases under future climate change, the global maize production in 2020 was used as the basis for dividing yield data into five components representing different yield levels according to various orders of magnitude. The global maize production was obtained from the Food and Agriculture Organization of the United Nations (FAO, https://www.fao.org/, accessed on 6 December 2022) a database which generates and shares updated information to ensure sustainable food production in agriculture, fisheries, and other areas.[41] These two sets of data were visualized using ArcGIS 10.8. Different colored points represent different diseases. The visualization graphic developed by this study showed that different diseases occur to different degrees around the world (Fig. 1).

In this study, we predicted the potential distribution of selected diseases under future climate conditions using bioclimatic projections based on four global climatic models (GCMs) (IPSL-CM6A-LR, MPI-ESM1-2-HR, MRI-ESM2-0, and UKESM1-0-LL), as well as three shared socioeconomic pathways (SSPs) (ssp126, SSP1 + representative concentration pathway (RCP)2.6; ssp245, SSP2 + RCP4.5; ssp585, SSP5 + RCP8.5) for three future periods (2050s average for 2041–2060, 2070s average for 2061–2080, and 2090s average for 2081–2100). Based on the general principles of fluid mechanics and thermodynamics, GCMs originated from numerical weather prediction, describing the dynamics of the atmosphere and oceans, providing a basis for past, present and future climate transitions, and estimating large-scale climate changes that may occur in the future.[42] In this study, four structurally independent models are selected, including two low climate sensitivity models (MPI-ESM1-2-HR, MRI-ESM2-0) and two high climate sensitivity models (IPSL-CM6A-LR, UKESM1-0-LL). At the same time, these models have comprehensive climate representativeness, from general representativeness (IPSL-CM6A-LR, MPIESM1-2-HR) to good representativeness (MRI-ESM2-0, UKESM1-0-LL).[43,44] RCPs are a series of integrated concentration and emission scenarios, which are used as input parameters for climate change prediction models under the influence of human activities in the 21st century to describe the emissions of greenhouse gases, reactive gases, aerosols, and the concentration of atmospheric components when future population, social economy, science and technology, energy consumption, and land-use change.[45] The RCPs include one high emission scenario (RCP8.5, baseline scenario without climate change policy

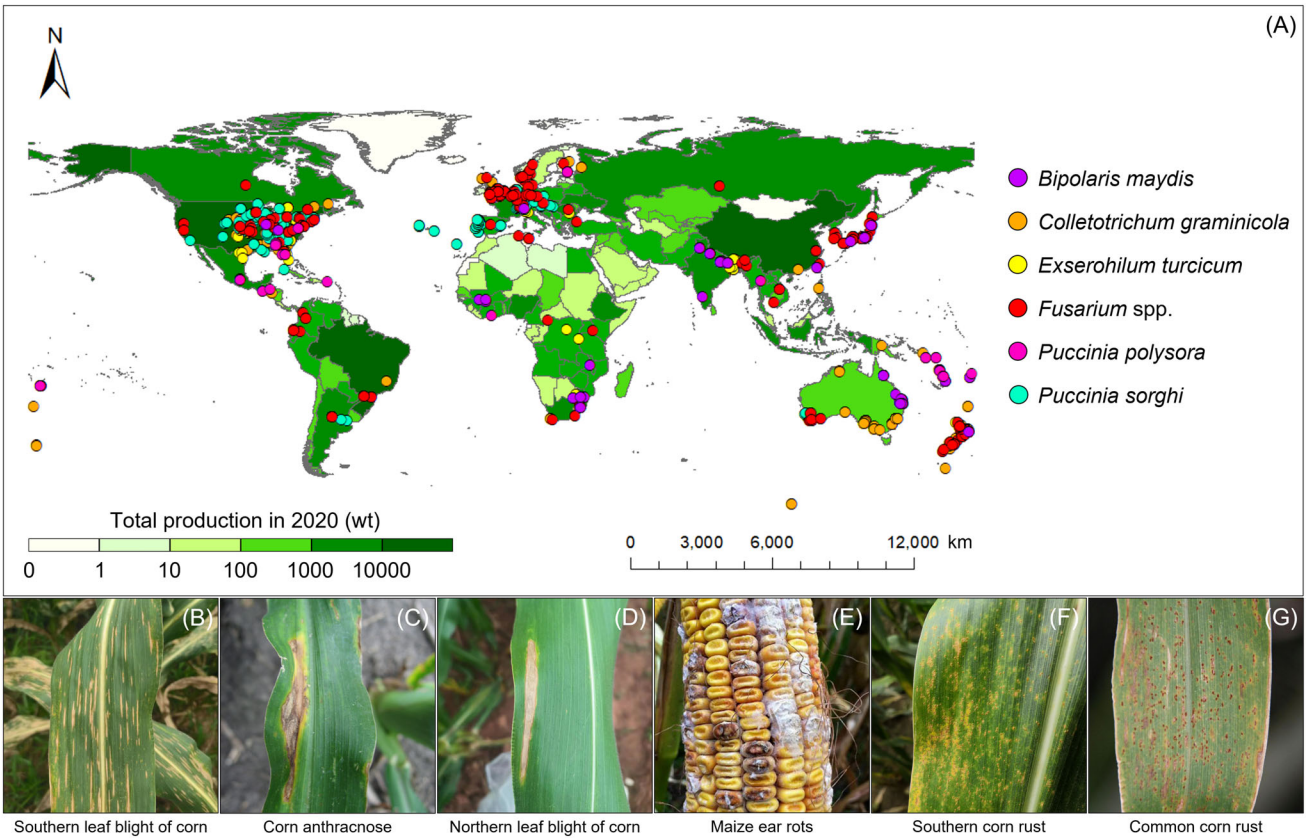


Figure 1. Recorded global maize production and distribution points of selected diseases in 2020, the symptoms of diseases were also presented. (A) Global maize production and diseases distribution in 2020. This research used two platforms, the Food and Agriculture Organization of the United Nations (FAO) and the Global Biodiversity Information Facility (GBIF), to obtain maize production and disease distribution, respectively. The data was visually emphasized by colored strips, indicating different orders of magnitude of yield, while colored dots pinpoint 757 instances of diseases. The pathogens causing six selected diseases are: (B) *Bipolaris maydis* (southern leaf blight of corn); (C) *Colletotrichum graminicola* (corn anthracnose); (D) *Exserohilum turcicum* (northern leaf blight of corn, NLB), in addition, the sexual pathogen *Setosphaeria turcica* is also the pathogen (sexual) causing NLB[40]; (E) the disease is represented by asexual pathogens; '*Fusarium* spp.' contains two pathogens (*F. graminearum* and *F. fujikuroi*) that cause the development of maize ear rots[40]; (F) *Puccinia polysora* (southern corn rust) and *Puccinia sorghi* (common corn rust).

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intervention), two medium emission scenarios (RCP4.5 and RCP6.0, climate scenarios under government intervention) and one low emission scenario (RCP2.6, scenario model of very low greenhouse gas concentrations),[46] here, this study chooses RCP4.5 in two medium emission scenarios, which is due to the implementation of afforestation policy and the increase of crop yield per unit area, while RCP4.5 is the emission mode of cultivated land area reduction.[47] Coupled Model Intercomparison Project (CMIP) is a project of the World Climate Research Program that provides climate predictions to understand past, present and future climate change. CMIP openly provides multi-model output in a standardized format and has organized six model comparison programs. CMIP and its associated data infrastructure have become key to the Intergovernmental Panel on Climate Change (IPCC) and other international and national climate assessments.[48] CMIP6 presents a more accurate simulation of extreme climate conditions, such as high temperatures and heavy rainfall, with results that closely match observations.[49] In CMIP6, RCP is updated to new SSPs. As an important part of the new generation of climate change scenarios, SSPs describes different trajectories of future socioeconomic system development and can reflect the relationship between socioeconomic development models and climate change risks.[50] It is pointed out that SSP126 is a scheme with very low greenhouse gas concentration, which requires emission reduction from the middle of this century and negative emission from 2080; the SSP585 is an emission scheme that increases at the current rate, which predicts a CO_2 concentration of 1135 ppm in 2100; between these two extreme scenarios, the SSP245 scenario represents the intermediate future scenario of partial social adaptation. In this study, SSP126, SSP245 and SSP585, which are strongly related to the occurrence of diseases, are selected as the research focus.[51–53] Data from CMIP6 were obtained from the WorldClim dataset (https://www.worldclim.org/, accessed on 25 April 2022), resulting in a total combination of 36 GCM-SSP-YEAR raster maps, all with a spatial resolution of 10 min (Supporting Information Table S1). The raster maps were stratified by R4.2.2 to obtain 19 bioclimatic variables (shown in abbreviation) and the projections were defined as GCS_WGS_1984 and exported in ASCII format for modeling use (the code used in this section is referenced in Supporting Information Note S1).

2.2 Model prediction

MaxEnt model (version 3.4.4) was used to identify the environmental variables affecting species distribution and simulate the potential geographic distribution of species.[54,55] The observed point data for the diseases was input into the 'Samples' section, while the layered climate data was input into the 'Environmental layers' section. The options to 'Create response curves', 'Make pictures of predictions' and 'Do jackknife to measure variable importance' were checked. Here, the 'Samples' represents a table containing the latitude and longitude of all disease occurrence points used in this study. The table type is '.csv'. The 'Environmental layers' represents 19 bioclimatic variables data for each GCM used in this study, and the data type is '.tiff'. The three selected options ('Create response curves', 'Make pictures of predictions' and 'Do jackknife to measure variable importance') represent the creation of response curves in the model prediction process to determine how each bioclimatic variable affects the prediction, the production of prediction result pictures to visualize the prediction results, and the jackknife test to determine the contribution rate of different environmental variables, respectively. The logistic output was utilized in MaxEnt model, which generated a continuous map with an estimated probability of presence ranging between 0 and 1. Of the distribution data loaded into the MaxEnt model, 75% were used for training and the remaining 25% for random testing. To reduce contingency and improve the accuracy of the results, ten-fold cross-validation was employed, where 25% of the existing points were randomly retained each time, thereby maximizing the use of existing data sets and dividing them multiple times.

The accuracy of the MaxEnt model was evaluated using the receiver operating characteristic (ROC) curve and the area under the curve (AUC) value. A higher AUC value indicates greater accuracy, with values ranging from 0.5 to 1. The evaluation criteria for AUC are poor (0.5–0.7), moderate (0.7–0.9), and excellent (0.9–1.0).[56,57] During the experiment, the MaxEnt model produced average AUC values ranging from 0.93 to 0.94 for the ten-fold cross-validation using different GCMs (Supporting Information Fig. S1), indicating that the MaxEnt model had a good predictive effect and is applicable for the analysis of disease prediction for selected diseases of maize. Additionally, the separate predictions for each disease showed AUC values close to 1 (Table S2), indicating the accuracy and reliability of using the model in predicting the future occurrence of each disease. The MaxEnt model employed an iterative algorithm to adjust individual evaluation factors and coefficients in the model predictions and calculate the contribution of environmental factors to the predictions.[58]

In this study, we obtained prediction maps of logistics output showing the potential distribution of selected diseases of maize under different climate scenarios in different years in the future. The MaxEnt model analysis outputs the probability of the distribution, ranging from 0 to 1, where a probability close to 1 indicates a higher probability of actual distribution. The median of the results obtained by ten-fold cross-validation was selected as the research object, and ArcMap 10.8 software was used to reclassify. The original raster pixel values 0 to 1 were assigned 1 to 5 at equal intervals for subsequent analysis and calculation (in the reclassified values 1 represents 1 for disease occurrence level, 2 for 3, 3 for 5, 4 for 7, and 5 for 9). World map in vector form was downloaded and the ratio was adjusted to 1:150 000 000. The reclassified raster map was symbolized using a color ramp according to the disease classification to visualize the prediction results. Additionally, the occurrence of different diseases was visualized, while the occurrence and superposition of diseases in three specific year intervals in the future were analyzed and compared.

2.3 Estimate of disease occurrence and yield loss

Previous study predicted the yield losses and pressure major crops face as a result of pests and diseases, indicating the proportion of crop yield reduction caused by different diseases in different regions and globally.[40] To estimate the degree of losses, the data provided from these studies were utilized and a quantified model to calculate the proportion of yield losses due to varying levels of morbidity. The occurrence of each disease was categorized into five levels (Fig. 2, Table S3), resulting in different percentages of yield loss (Table S4). Based on the levels, the total number of rasters was calculated by R4.2.2 and the rasters with different values in the reclassified graph were quantified (the code used in this section was referenced in Note S2).

To calculate the occurrence rate of different disease levels, the number of rasters corresponding to different disease levels after quantification is counted and divided by the total number of rasters to obtain the result. When the area change needs to be

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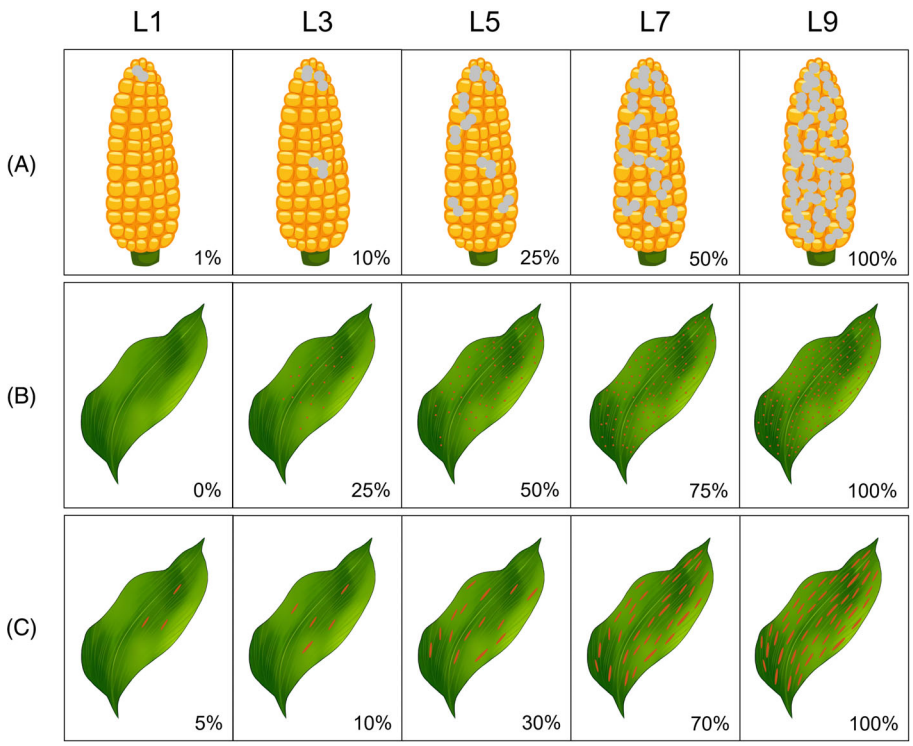


Figure 2. Schematic representation of occurrence levels of maize ear rots (A), southern/common corn rust (B) and other foliar disease (C), e.g., southern/northern leaf blight of corn and corn anthracnose, among them, northern leaf blight of corn caused grayish-brown lesions on the leaf surface. L1, L3, L5, L7 and L9 represented five different disease occurrence levels from light to heavy. The percentages marked in figures were the proportion of the occurrence area of the corresponding level on the host surface.

calculated, the percentage change is multiplied by the world area (except Antarctica) to obtain the area change of the disease.

To calculate the proportion of each level of raster to the total and determine the yield loss, the data were loaded into Excel and the following formula was used:

$$R_x = Q_x / Q_t \quad (1)$$

$$\text{Yield loss} = \sum (P \cdot Rx \cdot Dx)\ (x = 1,2,3,4,5) \quad (2)$$

where x represents the occurrence level, D is the proportion of yield reduction caused by different occurrence levels, R refers to the proportion of different occurrence levels in the future climate scenario and Q is the number of rasters. Based on the estimation of yield loss, the average yield loss caused by different diseases in different continents over the next 80 years was calculated (Table S5).

3 RESULTS

3.1 Bioclimatic variable contributions

The contribution of 19 bioclimatic variables used in the model prediction process varied. The results obtained from the ten-fold cross-validation method in the MaxEnt model were summarized to analyze the contribution of bioclimatic variables corresponding to each GCM (Fig. 3). The analysis showed that the major variables affecting the development of the selected diseases were Annual mean temperature (AMT), 27.5% (12.2% to 39.2%), Mean temperature of the coldest quarter (TCQ), 17.3% (1.5% to 30.4%), and Annual precipitation (AP), 14.9% (4.6% to 19.7%). Temperature and precipitation were found to be the main factors limiting the occurrence of the selected diseases. However, the analysis of single diseases indicated that the primary environmental variables affecting the frequency of pathogen occurrence was AMT, particularly for SCR (35.7%, 13.7% to 53.1%) (Table S6). Temperature was noted as the most limiting factor for the occurrence of maize diseases among the 19 bioclimatic variables.

3.2 Future occurrence predictions of selected diseases

The results showed that future maize diseases are concentrated in the United States, Europe, southern China and south-western South America, with a few occurrences in Australia and Africa under future climate scenarios (Fig. 4). Future changes (from the 2050s to the 2090s) in the frequency of these diseases are shown in Figs 4 and S2. According to this study's species potential distribution prediction using the IPSL-CM6A-LR model, disease extent showed an increasing trend over time, with the proportion of disease occurrence increasing from 18.94% to 19.45% (growth of approximately 674 781 km^2 of occurrence area) under the SSP126 scenario. The occurrence range displayed a U-shaped trend over time for both the SSP126 and SSP245 scenarios, with the greatest occurrence occurring in the 2050s at 18.94% and 18.77%, respectively. Conversely, the SSP585 scenario showed the least occurrence (18.77%, 18.21%, and 18.50%, respectively) (Fig. 4). Predictions using the MPI-ESM1-2-HR model were similar to the IPSL-CM6A-LR model, with the extent of disease occurrence increasing over time (approximately 1.614 million km^2) under the SSP126 scenario and the maximum disease area occurring at SSP245 (Fig. S2(1)). The MRI-ESM2-0 model analysis displayed an inverted U-shaped trend of disease extent over time for both the SSP126 and SSP585 scenarios, with the most severe occurrence in the 2070s (19.22% and 19.59% of disease extent, respectively). Levels 7 and 9 disease occurrences expanded under the SSP126 scenario, indicating a linear increase in disease severity

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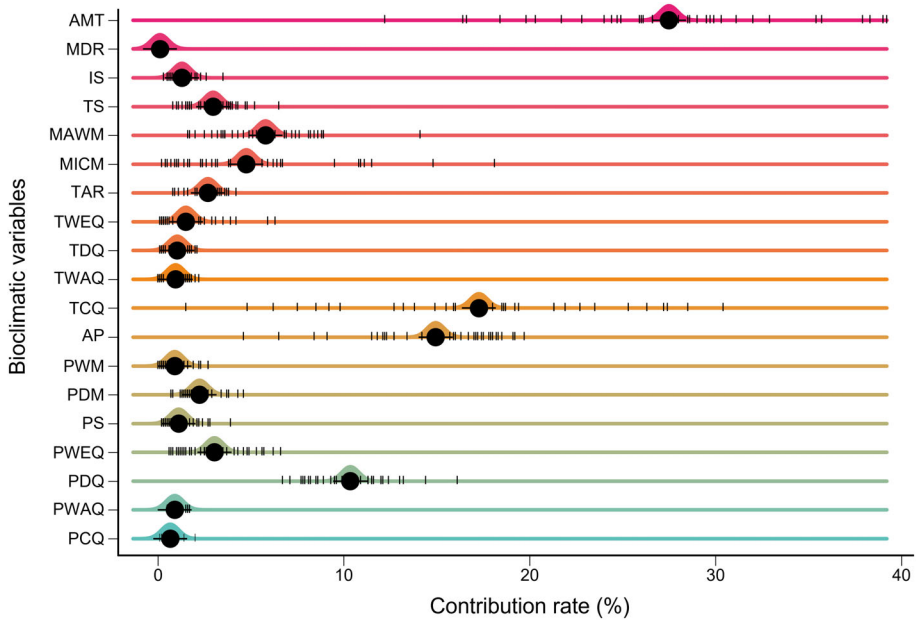


Figure 3. Contribution of each bioclimatic variable in the modeling of the future occurrence of selected diseases. Differences in the fitness effects of various bioclimatic variables on maize diseases were presented. The contribution rate of the bioclimatic variables to the disease occurrence are shown on the horizontal coordinates, while the different bioclimatic variables are represented on the vertical coordinates. The lines of different colors corresponding to the vertical coordinate represent different bioclimatic variables. The black vertical lines distributed on the colored lines represent the contribution rate of the bioclimatic variables to the occurrence of the disease after ten-fold cross-validation under different global climatic models (GCMs). The black dots and the protrusions of the lines are the mean value of the contribution rate of the bioclimatic variables. These variables included AMT (annual mean temperature, °C), MDR (mean diurnal range, °C), IS (isothermality, °C), TS (temperature seasonality, °C) MAWM (maximum temperature of the warmest month, °C), MICM (minimum temperature of the coldest month, °C), TAR (temperature annual range, °C), TWEQ (mean temperature of the wettest quarter, °C), TDQ (mean temperature of the driest quarter, °C), TWAQ (mean temperature of the warmest quarter, °C), TCQ (mean temperature of the coldest quarter, °C), AP (annual precipitation, mm), PWM (precipitation of the wettest month, mm), PDM (precipitation of the driest month, mm), PS (precipitation seasonality, mm), PWEQ (precipitation of the wettest quarter, mm), PDQ (precipitation of the driest quarter, mm), PWAQ (precipitation of the warmest quarter, mm), and PCQ (precipitation of the coldest quarter, mm).

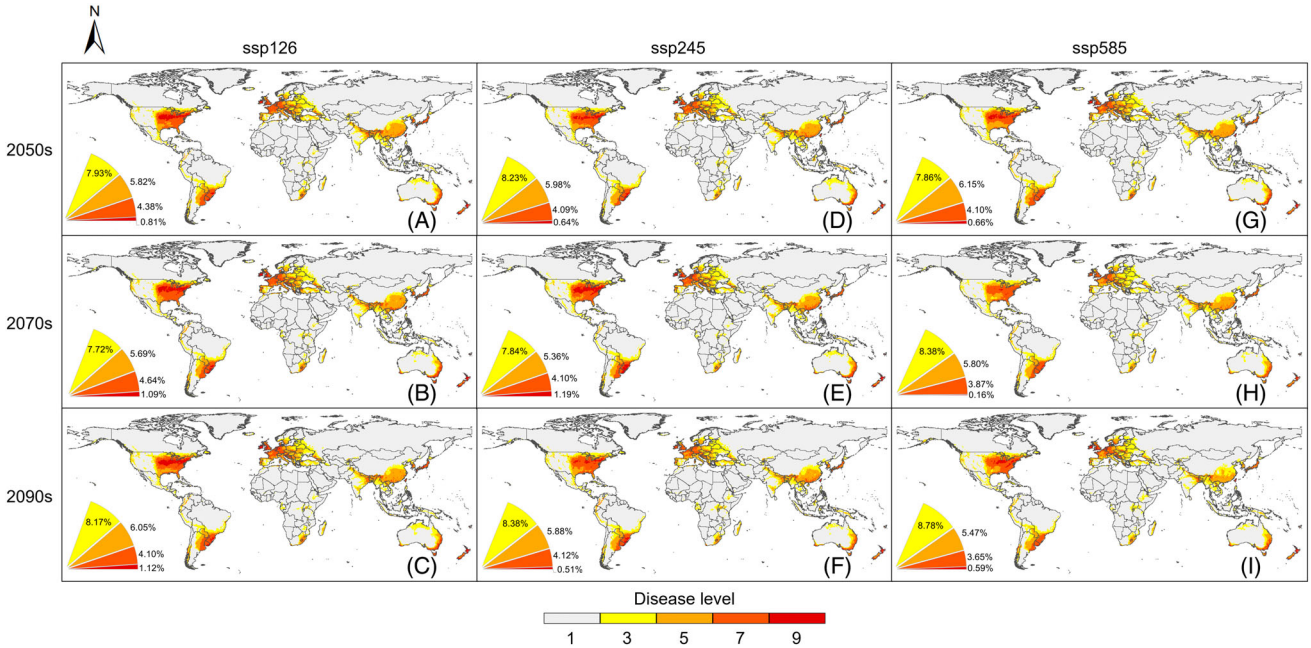


Figure 4. The comprehensive occurrence of six major maize diseases and the proportion corresponding to different disease levels of occurrence in each maize production regions predicted under the IPSL-CM6A-LR model in the future. Where 2050s, 2070s and 2090s represented 2041–2060, 2061–2080, and 2081–2100, respectively; ssp126 (SSP1 + RCP2.6), ssp245 (SSP2 + RCP4.5), and ssp585 (SSP5 + RCP8.5) represented the three different shared socioeconomic pathways; (A)–(I) ssp126_2041-2060, ssp126_2061-2080, ssp126_2081-2100, ssp245_2041-2060, ssp245_2061-2080, ssp245_2081-2100, ssp585_2041-2060, ssp585_2061-2080 and ssp585_2081-2100, respectively. The fan charts represented the proportion of different disease levels under the corresponding SSP-YEAR. The different colored bands represented the five different disease levels.

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despite non-linear changes in occurrence (Fig. S2(2)). Using the MaxEnt model under the UKESM1-0-LL model, the extent of occurrence under SSP245 and SSP585 was predicted to increase over time, with the extent of occurrence most severe in the 2090s, increasing by approximately 529 240 and 1 839 109 km^2, respectively (Fig. S2(3)).

3.3 Future occurrence predictions of SCR

In this study, the results indicated that SCR had the highest occurrence range (Fig. S3), followed by MER (Fig. S4), corn anthracnose (CA) (Fig. S5), NLB (Fig. S6), southern leaf blight of corn (SLB) (Fig. S7), and common corn rust (CCR) (Fig. S8). Interestingly, the occurrence of SCR was present in regions where all selected diseases were absent, including Australia, Africa, and South America. In these regions, the frequency of level 3 disease was significantly higher, accounting for nearly 28.50% of the five occurrence classes. In the remaining predicted results, the occurrence was less than 10%. Therefore, the total occurrence range was twice or more than that of the combined results. Additionally, South America and Australia exhibited a wide range of different occurrence levels, ranging from 3 to 9, indicating a high degree of occurrence. Among the four GCMs used in this study, the MRI-ESM2-0 model indicated the highest probability of disease occurrence, at 50.73% (ranging from 47.24% to 53.17%) (Fig. S3(3)). The IPSL-CM6A-LR model under the same SSPs exhibited a decreasing trend of occurrence, with the smallest occurrence area in the 2090s (Fig. S3(1)).

The changes in the potential suitable areas of selected diseases in the 2050s and 2090s under the SSP126 climatic scenario are shown in Fig. 5. Under this scenario, obviously, the areas of SCR increased the most widely (approximately 1 481 627 km^2), mainly in North America, central Africa, and south-western Asia, whereas the areas that decreased were mainly distributed in America, northeast Brazil and northwest Russia and Kazakhstan. For SLB, the extent of occurrence changed only in isolated areas. For NLB, CA, MER and CCR (approximately increased 7 856 884, 11 286 243, 1 405 722 and 5 395 956 km^2, respectively; decreased 50 361 767, 5 993 394, 81.7059 and 4 130 391 km^2, respectively), all of which are predicted to be present in Europe in the 2050s, occurrence varies among the different diseases by 2090. The occurrence of CA has expanded over varying areas on all continents except in Oceania, decreased occurrence in southern Asia. NLB has increased in North America, South America, western Europe, southern Asia, and northern Oceania, while there has been a marked decrease in central Europe. Increased occurrence of MER in North and South America, with a decrease in China and western South America. CCR is increasing in North America, Europe and Asia.

The response curves of environmental variables were also used to assess the probability of SCR occurrence, as exemplified by IPSL-CM6A-LR_ssp126 predicted results. In the 2050s, the bioclimatic variable that had the most significant impact on this disease was the AMT, and the frequency of SCR increased with the AMT until it reached 32.3 °C, at which point the SCR probability reached its maximum (Fig. S9(A)); In the 2070s and 2090s, the disease probability reached its maximum when the AMT was at 32.1 °C (Fig. S9(B),(C)).

3.4 Estimation of average yield loss for different GCMs

The results indicate that the overall yield loss caused by diseases from the 2050s to the 2090s would decrease under different SSPs, apart from the variation caused in the 2070s (Fig. 6). The yield loss caused in the 2090s was 9.28 × 10^4 t less than that of the 2050s under SSP126. However, the two interdecadal periods under SSP245 and SSP585 scenarios differ significantly, 2.38 × 10^5 t and 1.54 × 10^6 t, respectively. Despite the increase in yield loss caused by some diseases in specific scenarios in the 2070s, the highest impact of these selected diseases on maize yield loss over the next 80 years is likely to occur in the upcoming 2050s when the three scenarios are combined and averaged. However, even though the 2090s caused the smallest yield loss impact within the three intervals, the yield loss under each SSPs would still be higher than 3.5 × 10^7 t, indicating an increase of about 10^7 t compared to the 2020 yield reduction (https://www.fao.org/). Yield losses due to different diseases are presented in Fig. 6, with the length of the different colored bars representing the yield losses due to different diseases. The results revealed that the disease with the most significant impact on yield was SCR, accounting for 38.55% (36.05% to 39.50%) of the total yield losses, followed by MER, CA, NLB, CCR, and finally SLB. Combining the finding from the previous section, it can be inferred that, despite MER's regional extent being comparable to that of several other diseases (except SCR), MER has a severe influence on yield loss, accounting for 23.92% of the total yield losses and about 9.27 × 10^6 t of yield loss.

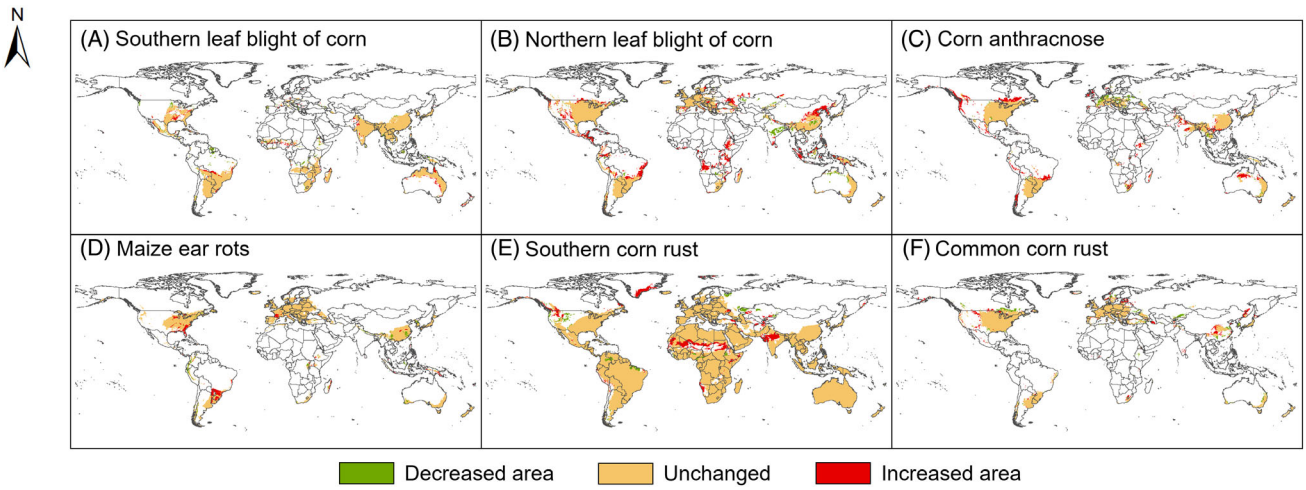


Figure 5. Changes from the 2050s to the 2090s in the potential suitable areas for six major maize diseases under IPSL-CM6A-LR_ssp126.

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3.5 Effect of selected diseases on maize yield on continents

The continents of the world differ in their major food crops and their acreage of maize cultivation. To better understand the negative impact of diseases on maize production on each continent, the possible yield losses under various climate scenarios in the 2050s were calculated. Using 2020 production data and continent-specific models, the effect of different diseases on maize yield was analyzed, using IPSL-CM6A-LR_ssp126 as an example. The findings showed that the largest yield loss is expected in the upcoming 2050s. The proportion of each disease in the 2050s under IPSL-CM6A-LR_ssp126 was selected and calculated based on the proportion of yield loss caused by each disease and the yield of each continent (Fig. 7). The yield loss of maize in the 2050s varied among different continents, with the largest yield reduction in North America (1.36×10^7 t), followed by Asia (1.21×10^7 t), South America, Europe, Africa, and finally Oceania. Notably, all the yield reductions caused by each disease had the most significant impact on North America. For diseases, SCR caused the largest yield loss, totaling 1.53×10^7 t across all continents.

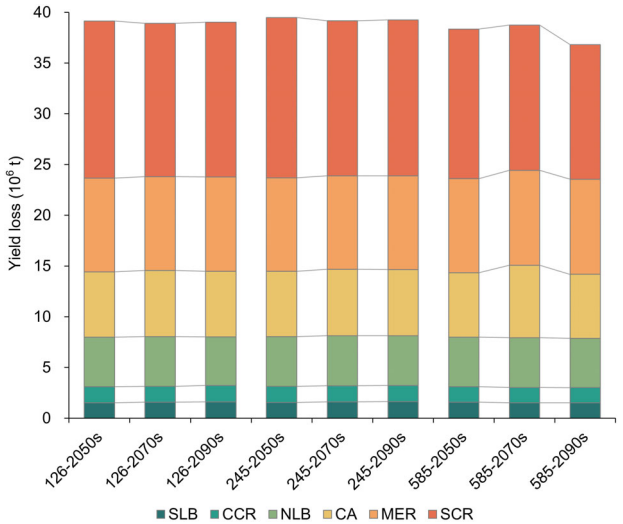


Figure 6. Calculation of average yield loss for different SSPs and year intervals under four global climatic models (GCMs). Where the different colored bars represent the yield loss caused by different kinds of diseases, the length of the bars represent the amount of yield loss caused, and the whole bars represent the total yield loss of the selected diseases under the same SSP-YEAR combination. 126-2050s, 126-2070s and 126-2090s represented SSP126_2041-2060, SSP126_2061-2080, and SSP126_2081-2100, respectively. In addition, 245-2050s, 245-2070s and 245-2090s represented SSP245_2041-2060, SSP245_2061-2080 and SSP245_2081-2100, respectively. Finally, 585-2050s, 585-2070s and 585-2090s represented SSP585_2041-2060, SSP585_2061-2080 and SSP585_2081-2100, respectively. SLB, CCR, NLB, CA, MER and SCR represented southern leaf blight of corn, common corn rust, northern leaf blight of corn, corn anthracnose, maize ear rots and southern corn rust, respectively.

4 DISCUSSION

Climate change has had a significant impact on agricultural productions, specifically affecting crop growth by reducing yield and quality. Indirectly, climate change also affects crop productivity by increasing the occurrence of pests and diseases, destabilizing ecosystems and exacerbating natural disasters, thus posing a threat to food security.[59] Previous studies have noted the negative impact of climate change on maize yield and of a single disease on maize growth.[27,60,61] In this study, the prevalence of selected diseases of maize affected by climate change was used to reflect the future trends in disease occurrence under climate change. By analyzing the frequency of disease occurrence, this study aimed to estimate the prospective yield loss caused by maize diseases under climate change. Ultimately, this research also provides targeted countermeasures to address the impact of diseases.

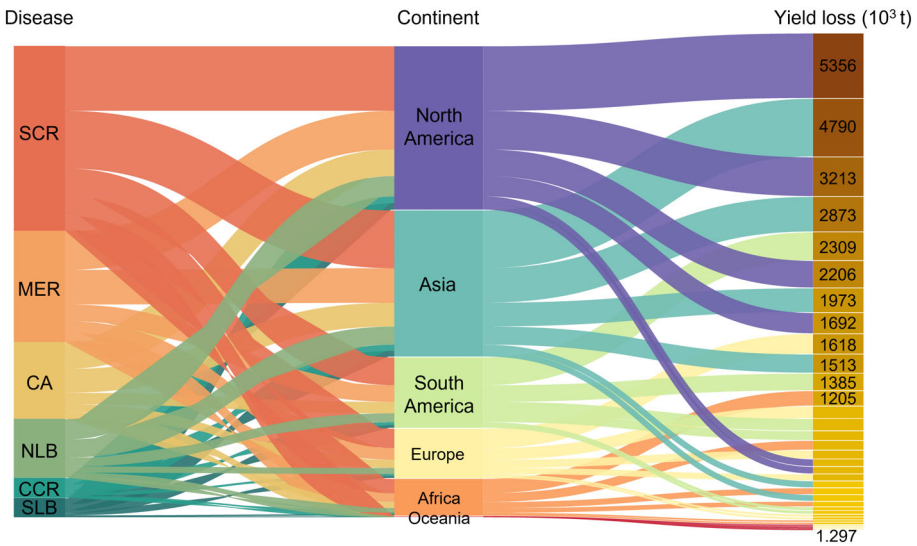


Figure 7. Maize yield losses on continents due to selected diseases under IPSL-CM6A-LR_ssp126_2041-2060. The first column represented different kinds of diseases, each indicated by a different color. The second column represented different continents, also indicated by various colors. The third column displayed a range of colors from light to dark, representing different degrees of yield loss, the unmarked parts indicate that the yield losses are less than 10^6 t. These parts are shown in the Supporting Information Table S4. The width of the line between each pair of columns reflected the amount of yield loss, which finally corresponded to each small column in the third column. SCR, MER, CA, NLB, CCR and SLB represented southern corn rust, maize ear rots, corn anthracnose, northern leaf blight of corn, common corn rust and southern leaf blight of corn, respectively.

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4.1 Uncertainty of future actual climate

Global temperatures are rising due to climate change, and drought is affecting crop production directly and indirectly.[62] According to the IPCC Sixth Assessment Report (AR6), the global surface temperature during 2011–2020 was 1.09°C higher than the temperature during 1850–1900, with human activity causing a 1.07°C increase.[63] Considering the current rise in global temperature that leads to drying up of rivers and a generally hotter and drier summer season, disease occurrence is directly influenced by these weather conditions.[64] This study thus focused on analyzing the impact of various environmental factors, such as temperature and precipitation on maize diseases by combining model outputs with results of regularized training gain. The results revealed that AMT, TCQ, and AP had the greatest influence on the distribution of maize diseases, while other environmental variables had relatively minor effects (Fig. 3). For instance, NLB and SLB require a relative humidity of 90% or higher to infection[65,66]; For MER, humidity conditions, particularly the amount of rainfall in summer, and prolonged rainy periods may lead to the growth of SCR and CCR.[67] Our findings highlight the importance of considering these environmental factors when studying the effects of climate change on crop production.

4.2 Multiple factors affecting pathogens

Various pathogens have unique optimal conditions for pathogenesis. With the changing climate, these pathogens exhibit different levels of adaptation.[68] The predicted regions of occurrence were approximately the same under different climate models (Figs 4 and S2(1),(3)), and the occurrence trends of different year interval corresponding to different SSPs of different climate models were not very consistent. For example, under the IPSL-CM6A-LR model, the overall occurrence frequency of the SSP126 scenario was increasing, while the frequency of occurrence in the other two climate scenarios was minimized in the 2050s and then increased (Fig. 4(A)–(C)). Differences were also observed in other model predictions. These differences can be attributed to a combination of factors, including the pathogen's receptivity to conditions, changes in environmental conditions, soil microorganisms, and the physiological regulation of the plant itself.[4–6,69–71] Additionally, human factors are emerging as important catalysts for disease occurrence, as human activities can influence pathogen occurrence by affecting climatic conditions.[72] Due to the complex interaction between multiple factors, the occurrence of pathogenic bacteria may not follow a certain pattern.

The results obtained from the MaxEnt model were used to calculate the number of rasters occupied by different disease classes, and to calculate the future occurrence frequency ratio by determining the proportion of rasters of different classes to the total number of rasters. The proportion of yield loss due to different kind of disease occurrence levels was calculated for various diseases by referring to the crop loss estimates for eight hotspots for global food security.[40] The yield in 2020 was used as the base data to calculate the yield loss due to diseases under different climate conditions in the future. Two high temperature-sensitive GCMs (IPSL-CM6A-LR, UKESM1-0-LL) and two low temperature-sensitive GCMs (MPI-ESM1-2-HR, MRI-ESM2-0) were utilized to predict the impact of climate change on yield loss. The results indicate that there are no consistent trends in the overall prediction results for yield loss caused by different GCMs under different SSPs. This may be due to the inevitable uncertainty in predictions based on a single global climate model.[58] Therefore, the average of the corresponding predictions under different GCMs under the same year interval of the same SSP was calculated, and the overall trend showed that the losses in each year interval under either SSP followed the same trend, reaching a maximum in the 2050s and at a minimum in the 2090s (Fig. 6). This demonstrated that multi-model forecasts can avoid the uncertainty of a single model and show trends that are more likely to occur in the future. We speculate that the global climate may no longer be suitable for the growth of pathogens by the 2090s due to the sustained high temperatures or droughts caused by these high temperatures in combination with human activities. Such environmental conditions would no longer be suitable for the rapid reproduction of pathogens.[73,74] However, the maximum amount of yield reduction occurred in the 2050s, as the temperature increased to reach the optimum temperature range for pathogen multiplication, and the growth of maize could not adapt to the temperature increase for a short time, resulting in an increase in yield reduction. Over time, the plants gradually adapted to the temperature change and the yield reduction decreased. For example, diseases like SLB, which prefer cold temperatures, would have their distribution range continuously reduced due to climate change. Therefore, the yield reduction caused solely by diseases would also gradually decrease.

4.3 The impact of the difference in planting area

After analyzing the yield loss caused by various diseases across different continents, it was found that SCR would cause the most detrimental loss in yield, followed by MER (Fig. 7). The severity of yield loss is closely related to the mode of transmission and the infection site of pathogens.[75] Given its long duration of cultivation, maize stalks left in the soil can promote infection in the next growing season.[76] Based on current production levels, North America has the highest maize production; therefore, this region would experience the most substantial yield loss (Fig. 7). This may be due to the high demand for maize leading to a decrease in spacing between rows, accelerating disease development in the leaves and base of stems, resulting in poor nutrition as well as reduced quality and quantity of maize.[77,78]

5 CONCLUSION

Through analyzing the occurrence of selected maize diseases and yield losses caused by them, the primary factors affecting disease development and crop yield were identified, which can inform strategies to combat these issues. However, it is crucial to note that other factors may also reduce maize yields, including pests and weeds, as well as extreme climatic conditions.[79] It is important to note that the estimated loss in yield is based on current levels of production and management control. Factors such as future levels of production and variability in means of control can significantly impact changes in maize yields. Therefore, there is a need for increased focus on prevention-oriented and integrated control measures in future maize production to composite the loss resulted from climate change. This includes strengthening plant quarantine protocols, employing scientific selection and breeding of highly resistant crop varieties, and utilizing molecular biology methods to develop disease-resistant varieties in a timely manner.[80,81] As global warming continues to increase, staggered sowing periods for maize could artificially create adverse conditions for disease that would help reduce the occurrence of its epidemics.[82,83] In addition, integrated pest management should be employed by diversifying control methods and coordinating their application economically, safely, and

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effectively. This is essential for achieving ecological, social, and economic sustainability, particularly for resource-limited farmers.[84] Elevation can sometimes be directly related to disease incidence.[85] Also, data like regional agricultural cultural practices (human factors) and economic assets of farmers (poverty) can be included as the 'Environmental layers'. The lack of nutrients such as nitrogen has also been reported to affect the occurrence of diseases.[86] However, these data are difficult to access, and these can be taken into account in future studies. More data are needed on disease occurrence to enhance prediction results and improve disease prevention and control strategies. By improving data collection, the prediction results could be enhanced to be more relevant, effective, and targeted for sustainable control of maize diseases.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no competing interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

AUTHOR CONTRIBUTIONS

Zihui Ma: acquisition and analysis of data, writing of the manuscript; **Wenbao Wang**: acquisition and analysis of data; **Xuanjing Chen**: assisted in data analysis and provided ideas; **Katrina Gehman**: revised the manuscript critically for important intellectual content; **Hua Yang**: provided article ideas; **Xuanjing Chen**: provided funds and ideas; **Yuheng Yang**: designed the experiments and revised the manuscript.

SUPPORTING INFORMATION

Supporting information may be found in the online version of this article.

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